

Debartha Paul

PhD Student in Statistics

Iowa State University · Ames, IA 50011

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EDUCATION

Doctor of Philosophy, Statistics

Aug'24 – Jul'29 (expected)

Iowa State University, Ames, IA

Advisor: Dr. Kyle Schindl

Master of Science, Statistics

Sep'22 – Jul'24

Presidency University, Kolkata, India

Bachelor of Science, Statistics (Hons.)

Aug'19 – Jun'22

Presidency University, Kolkata, India

RESEARCH

Research Interests: Causal Inference, Optimal Transport and Semiparametric Efficiency.

Manuscripts under Preparation:

- *A Geometric Framework for Distributional Causal Inference* with Kyle Schindl

TEACHING EXPERIENCE

Iowa State University

Aug'24 – May'25 & Aug'26 – Present

Graduate Teaching Assistant

- Grader, Statistical Methods III (STAT 5503), Fall'26.
- Lab instructor and grader, Intermediate Statistical Concepts and Methods (STAT 3201, formerly STAT 3010), Fall'26.
- Grader, Time Series Data Analysis (STAT 4720/5720), Spring'25.
- Lab instructor and grader, Intermediate Statistical Concepts and Methods (STAT 3010), Fall'24.

PROFESSIONAL EXPERIENCE

Corteva Agriscience

Aug'25 – Aug'26

Graduate Research Assistant (Remote)

- Identified an optimization bottleneck in an enterprise adaptive-experimentation pipeline where the algorithm reliably converged to a constrained local maximum rather than the global optimum.
- Resolved it by lifting the problem into a higher-dimensional space via **phantom variables**, adding degrees of freedom that let the optimizer escape the constrained maximum and close roughly 90% of the remaining gap to the best known solution.
- Built a Python benchmarking framework to evaluate the improved **Sequential Model-Based Optimization (SMBO)** against baselines across high-dimensional formulation spaces.
- Assessed convergence and stability through parallelized multi-seed simulations, using bootstrap confidence bands to quantify the performance gain over baseline.

MSA Labs

Apr'24 – Jun'24

Data Science Intern (Remote)

- Developed instructional materials teaching pharmacometrics practitioners to use Julia for statistical analysis.
- Created worked examples and tutorials covering statistical-computing workflows in Julia for a non-programming audience.

ACADEMIC PROJECTS

Probability-based Response-adaptive Allocation Design in Crossover Trials

Jun'24

*Masters dissertation at Presidency University, Kolkata. Supervised by **Dr. Saurav De***

- Proposed and developed an allocation design for crossover clinical trials that are both response-adaptive and probability-based.
- Used a simulation study to verify our allocation and also find equivalence tests for the treatments.

[GitHub](#)

Some Problems on Dice, Their Mathematical Treatments and Simulations

Nov'22

*Joint work with **Ahel Kundu***

- Derived closed-form solutions to ten classical dice problems using recursive expectation equations, inclusion-exclusion, and absorbing Markov chains.
- Computed expected hitting times via the fundamental matrix $(I - Q)^{-1}$ and solved an optimal-stopping problem for maximizing expected score.
- Verified each analytical result through Monte Carlo simulation in R, with reproducible seeds and convergence plots.

[GitHub](#) · [R-bloggers](#)

RELEVANT COURSEWORK

Theory: Measure Theory, Advanced Probability Theory

Methods: Linear Models and Design of Experiments, Advanced Statistical Modelling

Electives: Time Series Analysis, Advanced Bayesian Methods, Nonparametric Methods, Stochastic Processes in Modern Machine Learning

OPEN-SOURCE CONTRIBUTIONS

R Project Sprint, 2023 · [Link](#)

Aug'23

Contributor, base R (Remote)

- Selected to contribute to base R alongside R Core members.
- Contributed documentation fixes and translations for the language.

SimpleLinReg.jl · [GitHub](#)

2023

Built a Julia package end-to-end, including package structure, CI, test coverage, and documentation, as an exercise in the full development workflow.

SKILLS

Languages: R, Python, Julia, C, C++

Tools & Workflow: Git, $\LaTeX$, Quarto, Excel

Methods: Causal inference, Bayesian methods, optimization, time series, nonparametric inference

Spoken Languages: English, Bengali, Hindi

AWARDS & HONORS

International Youth Math Challenge
Finalist, Silver Honour in both editions.

2022 & 2023